

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE **COURSE CODE:** MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)
Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I G.Karthik Reddy-192525047 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Karthik Reddy

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

Q1. Smart Healthcare Diagnosis

The scenario states that Ravi has fever, cough, and body pain; fever+cough implies flu, flu+body pain implies viral infection, and viral infection requires medical attention.

(i) Facts, Entities, Predicates, Variables and Constants

Entity

- Patient: Ravi

Constants

- Ravi

Facts

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.

Predicates

- Fever(x)
- Cough(x)
- BodyPain(x)
- Flu(x)
- ViralInfection(x)
- MedicalAttention(x)

Variable

- x represents any patient.

(ii) Propositional Logic

Let:

- F = Ravi has fever
- C = Ravi has cough
- B = Ravi has body pain
- FL = Ravi has flu
- V = Ravi has viral infection
- M = Ravi requires medical attention

Facts

$F = \text{True}$

$C = \text{True}$

$B = \text{True}$

Rules

$F \wedge C \rightarrow FL$

$FL \wedge B \rightarrow V$

$V \rightarrow M$

Therefore:

$F \wedge C \rightarrow FL$

$FL \wedge B \rightarrow V$

$V \rightarrow M$

(iii) First-Order Logic

Facts

$\text{Fever}(\text{Ravi})$

$\text{Cough}(\text{Ravi})$

$\text{BodyPain}(\text{Ravi})$

Rules

$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$

$\forall x [(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]$

$\forall x [ViralInfection(x) \rightarrow MedicalAttention(x)]$

Here, $\forall x$ means “for every patient x.”

(iv) Unification

Unification 1

$\text{Fever}(x)$

$\text{Fever}(\text{Ravi})$

Substitution:

$\{x/\text{Ravi}\}$

Therefore:

$\text{Fever}(x) = \text{Fever}(\text{Ravi})$

Unification 2

$\text{Flu}(x)$

$\text{Flu}(\text{Ravi})$

Substitution:

$\{x/\text{Ravi}\}$

MGU = $\{x/\text{Ravi}\}$

(v) Resolution to Prove Viral Infection

We have the rule:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Convert to CNF:

$\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Second rule:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

CNF:

$\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$

Facts:

$\text{Fever}(\text{Ravi})$

$\text{Cough}(\text{Ravi})$

$\text{BodyPain}(\text{Ravi})$

To prove:

$\text{ViralInfection}(\text{Ravi})$

Negate the query:

$\neg \text{ViralInfection}(\text{Ravi})$

From:

$\text{Fever}(\text{Ravi})$

+

$\text{Cough}(\text{Ravi})$

we derive:

Flu(Ravi)

Then:

Flu(Ravi)

+

BodyPain(Ravi)

gives:

ViralInfection(Ravi)

This contradicts:

$\neg$ ViralInfection(Ravi)

Therefore:

ViralInfection(Ravi) is proved.

(vi) Complete Inference Sequence

Fever(Ravi)

+

Cough(Ravi)

↓

Flu(Ravi)

+

BodyPain(Ravi)

↓

ViralInfection(Ravi)

↓

MedicalAttention(Ravi)

Final Conclusion

Ravi has a viral infection and requires medical attention.

Q2. Student Academic Advising

The PDF states that Priya has attendance below 75% and has failed two courses. Low attendance requires attendance counseling, two or more failures require academic counseling, and students requiring both types need academic support.

(i) Knowledge Components

Entity

- Priya

Constants

- Priya

Facts

LowAttendance(Priya)

FailedTwoCourses(Priya)

Predicates

LowAttendance(x)

FailedTwoOrMore(x)

AttendanceCounseling(x)

AcademicCounseling(x)

AcademicSupport(x)

Rules

LowAttendance(x) $\rightarrow$ AttendanceCounseling(x)

FailedTwoOrMore(x) $\rightarrow$ AcademicCounseling(x)

AttendanceCounseling(x) $\wedge$ AcademicCounseling(x)

$\rightarrow$ AcademicSupport(x)

(ii) Propositional Logic

Let:

- L = Priya has low attendance
- F = Priya failed two courses
- AC = Priya requires attendance counseling
- CC = Priya requires academic counseling
- AS = Priya requires academic support

Facts

L

F

Rules

$L \rightarrow AC$

$F \rightarrow CC$

$AC \wedge CC \rightarrow AS$

(iii) First-Order Logic

$\forall x [LowAttendance(x) \rightarrow AttendanceCounseling(x)]$

$\forall x [FailedTwoOrMore(x) \rightarrow AcademicCounseling(x)]$

$\forall x [(AttendanceCounseling(x) \wedge AcademicCounseling(x))$
 $\rightarrow AcademicSupport(x)]$

Facts:

LowAttendance(Priya)

FailedTwoOrMore(Priya)

(iv) Unification and Substitution

Example 1

LowAttendance(x)

LowAttendance(Priya)

MGU:

$\{x/Priya\}$

Example 2

AcademicCounseling(x)

AcademicCounseling(Priya)

MGU:

$\{x/Priya\}$

(v) Resolution for Academic Support

Rules:

$LowAttendance(x) \rightarrow AttendanceCounseling(x)$

$\text{FailedTwoOrMore}(x) \rightarrow \text{AcademicCounseling}(x)$

$\text{AttendanceCounseling}(x) \wedge \text{AcademicCounseling}(x)$

$\rightarrow \text{AcademicSupport}(x)$

Facts:

$\text{LowAttendance}(\text{Priya})$

$\text{FailedTwoOrMore}(\text{Priya})$

Therefore:

$\text{AttendanceCounseling}(\text{Priya})$

and:

$\text{AcademicCounseling}(\text{Priya})$

Then:

$\text{AttendanceCounseling}(\text{Priya})$

+

$\text{AcademicCounseling}(\text{Priya})$

gives:

$\text{AcademicSupport}(\text{Priya})$

For resolution, negate the query:

$\neg \text{AcademicSupport}(\text{Priya})$

The rules and facts derive:

$\text{AcademicSupport}(\text{Priya})$

This creates a contradiction.

Therefore:

$\text{AcademicSupport}(\text{Priya})$ is proved.

(vi) Reasoning Steps

Low attendance

↓

Attendance counseling

↓

+

Academic counseling



Failed two courses



Academic support

Final Conclusion

Priya requires academic support because she needs both attendance counseling and academic counseling.

Q3. Intelligent Loan Approval

The PDF states that Arun has stable employment, sufficient income, high credit score, and good repayment history. The rules then derive preliminary approval, low-risk status, and loan recommendation.

(i) Entities, Facts, Predicates and Rules

Entity

Arun

Facts

StableEmployment(Arun)

SufficientIncome(Arun)

HighCreditScore(Arun)

GoodRepaymentHistory(Arun)

Predicates

StableEmployment(x)

SufficientIncome(x)

HighCreditScore(x)

GoodRepaymentHistory(x)

PreliminaryApproval(x)

LowRisk(x)

LoanApproved(x)

Rules

$\text{StableEmployment}(x) \wedge \text{SufficientIncome}(x)$

$\rightarrow \text{PreliminaryApproval}(x)$

$\text{HighCreditScore}(x) \wedge \text{PreliminaryApproval}(x)$

$\rightarrow \text{LowRisk}(x)$

$\text{LowRisk}(x) \wedge \text{GoodRepaymentHistory}(x)$

$\rightarrow \text{LoanApproved}(x)$

(ii) Propositional Logic

Let:

- E = Arun has stable employment
- I = Arun has sufficient income
- C = Arun has high credit score
- R = Arun has good repayment history
- P = Arun gets preliminary approval
- L = Arun is low-risk
- A = Arun is approved for loan

Facts

E

I

C

R

Rules

$E \wedge I \rightarrow P$

$C \wedge P \rightarrow L$

$L \wedge R \rightarrow A$

(iii) FOL Representation

$\forall x [(\text{StableEmployment}(x) \wedge \text{SufficientIncome}(x))$

$\rightarrow \text{PreliminaryApproval}(x)]$

$\forall x [(HighCreditScore(x) \wedge PreliminaryApproval(x))$
 $\rightarrow LowRisk(x)]$

$\forall x [(LowRisk(x) \wedge GoodRepaymentHistory(x))$
 $\rightarrow LoanApproved(x)]$

(iv) Unification and Substitution

Unification 1

StableEmployment(x)

StableEmployment(Arun)

Substitution:

$\{x/Arun\}$

Unification 2

LoanApproved(x)

LoanApproved(Arun)

Substitution:

$\{x/Arun\}$

MGU = $\{x/Arun\}$

(v) Resolution for LoanApproved(Arun)

Rules in CNF:

$\neg StableEmployment(x)$

$\vee \neg SufficientIncome(x)$

$\vee PreliminaryApproval(x)$

$\neg HighCreditScore(x)$

$\vee \neg PreliminaryApproval(x)$

$\vee LowRisk(x)$

$\neg LowRisk(x)$

$\vee \neg GoodRepaymentHistory(x)$

$\vee LoanApproved(x)$

Facts:

StableEmployment(Arun)

SufficientIncome(Arun)

HighCreditScore(Arun)

GoodRepaymentHistory(Arun)

From the first rule:

PreliminaryApproval(Arun)

From the second rule:

LowRisk(Arun)

From the third rule:

LoanApproved(Arun)

produces a contradiction.

Therefore:

LoanApproved(Arun) is proved.

(vi) Final Inference

Stable Employment

+

Sufficient Income

↓

Preliminary Approval

+

High Credit Score

↓

Low Risk

+

Good Repayment History

↓

Loan Approved

Final Conclusion

Arun can be recommended for loan approval.

Q4. Vehicle Fault Diagnosis

The PDF states that Car101 does not start and has dim headlights. These conditions imply a battery problem; a battery problem requires inspection, and inspection leads to service marking.

(i) Facts and Rules

Facts

DoesNotStart(Car101)

DimHeadlights(Car101)

Rules

DoesNotStart(x) $\wedge$ DimHeadlights(x)

$\rightarrow$ BatteryProblem(x)

BatteryProblem(x)

$\rightarrow$ BatteryInspection(x)

BatteryInspection(x)

$\rightarrow$ MarkedForService(x)

(ii) Propositional Logic

Let:

- D = Car101 does not start
- H = Car101 has dim headlights
- B = Car101 has battery problem
- I = Car101 requires battery inspection
- S = Car101 is marked for service

Facts

D

H

Rules

$D \wedge H \rightarrow B$

$B \rightarrow I$

$I \rightarrow S$

(iii) FOL Predicates and Rules

Predicates:

DoesNotStart(x)

DimHeadlights(x)

BatteryProblem(x)

BatteryInspection(x)

MarkedForService(x)

Rules:

$\forall x [(DoesNotStart(x) \wedge DimHeadlights(x))$
 $\rightarrow BatteryProblem(x)]$

$\forall x [BatteryProblem(x) \rightarrow BatteryInspection(x)]$

$\forall x [BatteryInspection(x) \rightarrow MarkedForService(x)]$

(iv) Unification

Example 1

DoesNotStart(x)

DoesNotStart(Car101)

Substitution:

$\{x/Car101\}$

Example 2

BatteryProblem(x)

BatteryProblem(Car101)

Substitution:

$\{x/Car101\}$

Therefore:

MGU = $\{x/Car101\}$

(v) Resolution for Battery Problem

Rule:

$DoesNotStart(x) \wedge DimHeadlights(x)$

$\rightarrow BatteryProblem(x)$

CNF:

$\neg \text{DoesNotStart}(x)$

$\vee \neg \text{DimHeadlights}(x)$

$\vee \text{BatteryProblem}(x)$

Facts:

$\text{DoesNotStart}(\text{Car101})$

$\text{DimHeadlights}(\text{Car101})$

Negated query:

$\neg \text{BatteryProblem}(\text{Car101})$

Using resolution:

$\text{DoesNotStart}(\text{Car101})$

+

$\text{DimHeadlights}(\text{Car101})$

gives:

$\text{BatteryProblem}(\text{Car101})$

This contradicts:

$\neg \text{BatteryProblem}(\text{Car101})$

Therefore:

$\text{BatteryProblem}(\text{Car101})$ is proved.

(vi) Service Inference

From:

$\text{BatteryProblem}(\text{Car101})$

we derive:

$\text{BatteryInspection}(\text{Car101})$

Then:

$\text{BatteryInspection}(\text{Car101})$

gives:

$\text{MarkedForService}(\text{Car101})$

Complete Inference

Does Not Start

+

Dim Headlights

↓

Battery Problem

↓

Battery Inspection

↓

Marked for Service

Final Conclusion

Car101 has a battery problem and should be marked for service.

Q5. Cybersecurity Threat Detection

The PDF states that User101 has five failed login attempts and an unusual login location. Multiple failed logins imply a possible brute-force attack; brute-force plus unusual location implies suspicious activity; suspicious activity plus unauthorized confidential-file access implies a possible security breach.

(i) Facts, Entity, Predicates and Rules

Entity

User101

Facts

FiveFailedLogins(User101)

UnusualLocation(User101)

The scenario also uses:

UnauthorizedFileAccess(User101)

to derive the final breach.

Predicates

FiveFailedLogins(x)

UnusualLocation(x)

BruteForceAttack(x)

SuspiciousActivity(x)

UnauthorizedFileAccess(x)

SecurityBreach(x)

Rules

FiveFailedLogins(x)

→ BruteForceAttack(x)

BruteForceAttack(x) ∧ UnusualLocation(x)

→ SuspiciousActivity(x)

SuspiciousActivity(x) ∧ UnauthorizedFileAccess(x)

→ SecurityBreach(x)

(ii) Propositional Logic

Let:

- F = User101 has five failed logins
- U = User101 logged in from unusual location
- B = User101 has brute-force attack
- S = User101 has suspicious activity
- A = User101 has unauthorized file access
- R = User101 has possible security breach

Facts

F

U

A

Rules

$F \rightarrow B, B \wedge U \rightarrow S \ \& \ S \wedge A \rightarrow R$

(iii) FOL Rules

$\forall x [\text{FiveFailedLogins}(x)$

$\rightarrow \text{BruteForceAttack}(x)]$

$\forall x [(\text{BruteForceAttack}(x) \wedge \text{UnusualLocation}(x))$

$\rightarrow \text{SuspiciousActivity}(x)]$

$\forall x [(\text{SuspiciousActivity}(x) \wedge \text{UnauthorizedFileAccess}(x))$

$\rightarrow \text{SecurityBreach}(x)]$

(iv) Unification and Substitution

Unification 1

$\text{FiveFailedLogins}(x)$

$\text{FiveFailedLogins}(\text{User101})$

Substitution:

$\{x/\text{User101}\}$

Unification 2

$\text{SuspiciousActivity}(x)$

$\text{SuspiciousActivity}(\text{User101})$

Substitution:

$\{x/\text{User101}\}$

MGU = $\{x/\text{User101}\}$

(v) Resolution for Security Breach

Rules in CNF:

$\neg \text{FiveFailedLogins}(x)$

$\vee \text{BruteForceAttack}(x)$

$\neg \text{BruteForceAttack}(x)$

$\vee \neg \text{UnusualLocation}(x)$

$\vee \text{SuspiciousActivity}(x)$

$\neg \text{SuspiciousActivity}(x)$

$\vee \neg \text{UnauthorizedFileAccess}(x)$

$\vee \text{SecurityBreach}(x)$

Facts:

FiveFailedLogins(User101)

UnusualLocation(User101)

UnauthorizedFileAccess(User101)

From the first rule:

BruteForceAttack(User101)

From the second:

SuspiciousActivity(User101)

From the third:

SecurityBreach(User101)

Negated query:

$\neg$ SecurityBreach(User101)

Resolution gives a contradiction.

Therefore:

SecurityBreach(User101) is proved.

(vi) Complete Reasoning Chain

Five Failed Login Attempts

↓

Brute-Force Attack

+

Unusual Location

↓

Suspicious Activity

+

Unauthorized File Access

↓

Possible Security Breach

Final Conclusion

User101 is involved in a possible security breach.

